Spring 2026

MIST 6380: Advanced Data Management and Analytics

Department of Management Information Systems
Terry College of Business

Course Information

CRN	48072
Class Schedule	Tue/Thu 8:15 am – 9:35 am
Classroom	Correll Hall 313
Prerequisites	MIST 7600 or MIST 4610 or MIST 4610E

Instructor Information

Instructor:	Jiyong Park, Ph.D.
Office:	Benson Hall C428
Office Hours:	Thursday 12:30 pm – 2:00 pm, or by appointment
E-Mail:	jiyong.park@uga.edu

Note: *The course syllabus is a general plan for the course; deviations announced to the class by the instructor may be necessary.*

Course Description

Provides students with knowledge of emerging technical issues related to data science, along with experience of diverse methods and technologies related to all aspects of data science, from data collection to optimization.

Business analytics involves the science of analyzing large quantities of data to discover meaningful patterns and generate insights as well as the art of data exploration, visualization, and communication to support decision-making and create business value. This course provides students with advanced knowledge, methods, and techniques of data management and analytics.

Learning Outcomes

Upon completing this course, students will have first-hand experiences and skills in the use of analytics tools and build their analytical capabilities to utilize data for decision-making. Students will learn:

- Basics of Python language
- Python packages and libraries for analytics
- Data manipulation and handling
- Data organization, integration, and transformation
- Connecting to databases in Python
- Working SQL and NoSQL in Python

- Data collection from websites
- Using Application Programming Interfaces (APIs) for third-party integration
- Working with AI tools for data management and analytics
- Apply critical thinking to real-world problems and design effective applications to solve them

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Course Materials

All information and materials pertaining to this course will be available at eLearning Commons (eLC). Students will be responsible for any information and announcements posted at eLC. All graded deliverables will be submitted electronically through eLC.

Textbook

There is no required textbook. Although additional learning materials and reading lists may be provided each week (free of charge), this course is primarily based on lecture notes and codes provided by the instructor. Supplementary eBooks for this course include:

- A Whirlwind Tour of Python [WTOP]: Fast-paced introduction to essential features of Python
- Python Data Science Handbook [PDSH]: Handbook for modern Python tools for data analytics

Software

This course makes extensive use of **Python**. Jupyter Notebook will be used as the primary development environment. Students need to be prepared with the required software and bring their personal laptops to the classroom. There are two options.

- Option 1: You can install Anaconda (for Python) and then Jupyter Notebook on your computer. It is what the instructor will use in the classroom.
- Option 2: You can use the Google Colab that provides a web interface for Jupyter Notebook. It's easy to access and collaborate with other people. While charged plans to unlock advanced features and more computing resources are available, there is no need for advanced features in this course (free version is okay).

Policy on AI Tools such as ChatGPT

This course focuses on developing programming, data management, and analytics skills. The use of generative artificial intelligence (AI) tools, such as ChatGPT, can hinder learners' ability to acquire these skills. Additionally, correcting and improving the output from ChatGPT or similar tools requires sufficient knowledge; otherwise, it may be more time-consuming and challenging than completing the work independently. Therefore, unless explicitly permitted, the use of ChatGPT or similar tools in course activities will be considered an academic honesty violation and is strictly prohibited. Unauthorized or improper use of such tools will constitute a breach of academic integrity.

That said, it is important to recognize the transformative potential of AI in data management and analytics beyond simple code generation. Learning how to work effectively with AI to maximize its potential is increasingly crucial. To this end, the course will introduce purposeful ways to leverage AI tools for analytics and programming, including large language models and other generative AI systems (e.g., debugging with GenAI and establishing the credibility of GenAI applications through techniques such as prompt engineering and retrieval-augmented generation). These sessions aim to help students learn to use AI as a

complement and assistant to their learning, and ultimately their careers, rather than as a substitute for critical thinking and problem-solving.

Grading Policy

Your final points will be converted to a letter grade using the following criteria. There will also be extra credit outside of the regular points.

Item	Points
Weekly Quizzes	100
Individual Assignments	400
Group Project	200
Midterm Exam	100
Final Exam	200
TOTAL	1000
+ Extra Credit (Optional)	100

Points	Letter Grade
930 and above	A
900 – 929.99	A-
870 – 899.99	B+
830 – 869.99	B
800 – 829.99	B-
770 – 799.99	C+
730 – 769.99	C
700 – 729.99	C-
600 – 699.99	D
599.99 and below	F

Weekly Quizzes (Points 100)

There will be 10 weekly quizzes throughout the semester. Every week's lectures will be followed by short questions that are designed to test understanding of lecture materials.

Individual Assignments (Points 400)

There will be several individual assignments throughout the semester. Individual activities are *individual activities*. Help from within the class or outside will be considered cheating. When using external resources and materials (e.g., codes), a proper citation of the original material is required. Please check with the instructor if you have any doubts or questions about what constitutes acceptable use of help and external resources.

Group Project (Points 200)

There will be a group project with teams of two to three students throughout the semester, since learning with peers and working collaboratively is an effective way to learn. Only students who are permitted by the instructor in advance can do solo work. The group shares the grades of the group project. Note that the goal of these activities is to work together and not to split or distribute the work. It is expected that every group member to have a perfect understanding of every component of their project, as the concepts required to complete the final project will also be tested in individual exams.

Midterm Exams (Points 100)

There will be a midterm exam during the regular class. The exams will test knowledge and skills of data management and analytics and the use of software, which is similar to homework assignments.

Final Exam (Points 200)

The final exam is a longer 3-hour exam scheduled according to the university calendar and contains multiple-choice questions similar to quizzes and programming questions similar to homework assignments.

Extra Credit (Optional) (Extra Points 100)

Extra credit will be granted upon completion of online courses for MIS or data analytics. Detailed guidance will be provided during the course.

[Tentative] Class Schedule

Week	Date	Module	Topic	Supplementary Readings
1	Jan. 13		Course Introduction	
	Jan. 15	Python Fundamentals	Python Language Basics I	WTOP Ch. 2–8
2	Jan. 20		Python Language Basics II	WTOP Ch. 9–15
	Jan. 22		Data Indexing and Selection	PDSH 3.1, 3.2, 3.5
3	Jan. 27		Operating on Data I	PDSH 3.3
	Jan. 29		Operating on Data II	PDSH 3.4, 3.8
4	Feb. 3		Combining Datasets	PDSH 3.6, 3.7
	Feb. 5		Working with String and Time Series	PDSH 3.10, 3.11
5	Feb. 10	Advanced Data Management	Web Scrapping I	
	Feb. 12		Web Scrapping II	
6	Feb. 17		Using APIs in Python	
	Feb. 19		Review of SQL & NoSQL	
7	Feb. 24		Working SQL in Python I (MySQL)	
	Feb. 25		Working SQL in Python II (MySQL)	
8	Mar. 3		Working NoSQL in Python I (MongoDB)	
	Mar. 5		Working NoSQL in Python II (MongoDB)	
9	Mar. 10/12		Spring Break (No Class)	
10	Mar. 17		Midterm Exam	
	Mar. 19		Project Workday	
11	Mar. 24	Advanced Data Analytics	Introduction to Large Language Model (LLM) and Generative AI (GenAI)	
	Mar. 26		Proprietary vs. Open-Source Models	
12	Mar. 31		Prompt Engineering	
	Apr. 2		Fine-Tuning	
13	Apr. 7		LangChain & LangGraph I	
	Apr. 9		LangChain & LangGraph II	
14	Apr. 14		Retrieval-Augmented Generation (RAG) I	
	Apr. 16		Retrieval-Augmented Generation (RAG) II	
15	Apr. 21		Conversational Agents I	
	Apr. 23		Conversational Agents II	
16	Apr. 30		Final Exam	

- **The following schedule provides general guidelines and is subject to change.** A detailed schedule prior to the beginning of each week will be posted on eLC. It is the student's responsibility to stay on track with course materials on a weekly basis to be successful in the course.
- Deadlines of course deliverables will be posted to eLC. Make sure you check eLC calendar and note down these deadlines.
- Check UGA Academic Calendar for other important dates (e.g., registration, drop/add, withdrawal, and final exam dates).

Communication Policy

The best way to communicate with the instructor is to take advantage of office hours. If you wish to meet with the instructor outside of regular office hours, feel free to email to make an appointment.

If your question is about a topic that others can benefit from and that does not involve revealing a large portion of personal work, please direct it to the discussion forum on eLC. If your question requires sharing your personal work, or if you want to discuss a personal issue, you can email the instructor.

Attendance and Participation Policy

Attendance is not required for this course but highly encouraged. As course materials are designed in a cumulative manner, it is nearly impossible to catch up if you do not follow the first couple of classes. All materials that are needed for the assignments and exams will be covered in the classroom, so class attendance is critical to your success in this course.

Late Submission and Makeup Exam Policy

Keep in mind due dates of all course activities posted on eLC. Assignments and quizzes submitted after due dates will receive a penalty of a 20% deduction for each day late. Any submissions after 5 days from due dates will not be accepted unless special accommodations are permitted by the instructor in advance.

In general, there are no makeup exams. A makeup exam is possible only when the student misses the exam due to extenuating circumstances. In this situation, you are required to submit a letter from Student Care and Outreach to support this request. Any obstacles preventing the student from testing during the scheduled window must be communicated with the instructor via email 48 hours prior to the exam start-time to request accommodations. If a student must miss an exam with a valid excuse, the final will count as a makeup exam.

Special Accommodation

For any special accommodation, especially for makeup examinations or extension of deadlines, you should inform the instructor ahead of time (ASAP). Proper documentation is needed to request special accommodation. For illness and health issues, a medical note is required. A last-minute notice or undocumented request will not be accepted.

Academic Honesty

UGA Student Honor Code: "*I will be academically honest in all of my academic work and will not tolerate academic dishonesty of others*."

A Culture of Honesty, the University's policy and procedures for handling cases of suspected dishonesty, can be found at honesty.uga.edu.

As a University of Georgia student, you have agreed to abide by the University's academic honesty policy, "A Culture of Honesty," and the Student Honor Code. All academic work must meet the standards described in "A Culture of Honesty." Lack of knowledge of the academic honesty policy is *not* a reasonable explanation for a violation. Questions related to course assignments and the academic honesty policy should be directed to the instructor.

UGA Well-being Resources: https://well-being.uga.edu

UGA Well-being Resources promote student success by cultivating a culture that supports a more active, healthy, and engaged student community.

- Anyone needing assistance is encouraged to contact Student Care & Outreach (SCO) in the Division of Student Affairs at 706-542-8479 or visit sco.uga.edu. Student Care & Outreach helps students navigate difficult circumstances by connecting them with the most appropriate resources or services. They also administer the Embark@UGA program which supports students experiencing, or who have experienced, homelessness, foster care, or housing insecurity.

UGA provides both clinical and non-clinical options to support student well-being and mental health, any time, any place. Whether on campus, or studying from home or abroad, UGA Well-being Resources are here to help.

- Well-being Resources: *https://well-being.uga.edu*.
- Student Care and Outreach: https://sco.uga.edu
- University Health Center: https://healthcenter.uga.edu
- Counseling and Psychiatric Services: https://caps.uga.edu or CAPS 24/7 crisis support at 706-542-2273
- Health Promotion/Fontaine Center: https://healthpromotion.uga.edu
- Disability Resource Center and Testing Services: https://drc.uga.edu

Additional information, including free digital well-being resources, can be accessed through the UGA app or by visiting https://well-being.uga.edu.

Inclusive Excellence

The Terry College of Business is committed to promoting an inclusive learning and working environment among its students, faculty, and staff. This class welcomes the open exchange of ideas and values freedom of thought and expression and provides a professional environment that recognizes the inherent worth of every person. It aims to foster dignity, understanding, and mutual respect among all individuals in the class.